

A VISUAL, WORKED-EXAMPLE GUIDE

# Implied Orders

How a matching engine builds synthetic liquidity from the orders that already exist — explained from the ground up.

Implied IN

Implied OUT

No new liquidity

Est. reading: a full session · Level: beginner

*J. Persson, 2026*

START HERE

# What you'll walk away understanding

01

## The order book

How buy and sell interest is stored and matched.

02

## Spreads & combos

Instruments built from two or more other instruments.

03

## Implied IN

Turning outright leg orders into a spread quote.

04

## Implied OUT

Turning a spread + a leg into an outright quote.

05

## Why it's powerful

Tighter markets and more fills — for free.

06

## The catch

No new liquidity, and why depth must be capped.

PRIMER

# Dated instruments & combos

Why oil trades by delivery month — and how a single combo order bundles two months into one.

# A dated instrument settles in a specific month

An oil future isn't just "oil" — it's a barrel for delivery in a named month. Each month is a separate contract with its own price and its own order book.

|                                          |                                          |                                          |                                          |                                          |                                          |
|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|
| <b>FEB</b><br>WTI Crude<br><b>\$77.8</b> | <b>MAR</b><br>WTI Crude<br><b>\$78.2</b> | <b>APR</b><br>WTI Crude<br><b>\$78.5</b> | <b>MAY</b><br>WTI Crude<br><b>\$78.8</b> | <b>JUN</b><br>WTI Crude<br><b>\$79.1</b> | <b>JUL</b><br>WTI Crude<br><b>\$79.3</b> |
|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|

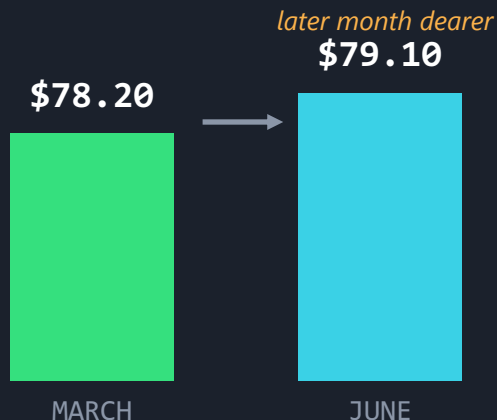
*Illustrative WTI prices, \$/barrel.*

"March oil" and "June oil" are two different instruments — you can buy one and sell the other independently.

# Why March and June don't cost the same

The price gap between months reflects the cost and expectations of holding oil over time.

## THE SAME BARREL, TWO DATES



## WHAT DRIVES THE GAP

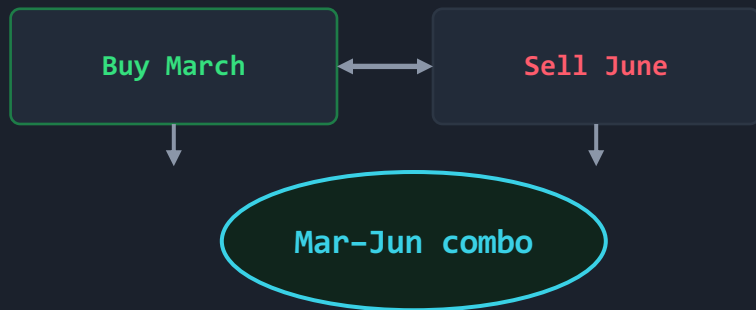
- **Storage & insurance**  
holding barrels until June costs money
- **Financing**  
cash tied up in the barrel earns interest elsewhere
- **Seasonality & outlook**  
expected supply and demand shift by month

June > March → “contango”

# A combo order trades two months at once

Instead of sending two separate orders, a combo (spread) order bundles both legs into ONE instrument that fills together, in a fixed 1:1 ratio.

## TWO LEGS, ONE TICKET



## HOW IT'S PRICED

**combo price = March - June**

One number to quote, one fill for both legs.

- No "legging risk" — you never get one leg without the other.
- You trade the relationship between months, not oil's outright direction.

# Worked example: buying the Mar–Jun combo

You buy 1 Mar–Jun combo. With March at 78.20 and June at 79.10, the two legs fill together:

**A Buy the March leg**

fill @ 78.20

**B Sell the June leg**

fill @ 79.10

**=** combo price =  $78.20 - 79.10 = -0.90$

net cash =  $+79.10 - 78.20 = +0.90$  credit

YOU PAID

**-0.90**

*for the spread*

You are now LONG the March–June time spread — one fill, both legs, no legging risk.

*Illustrative prices, \$/barrel.*

# Closing the trade: how you take profit

You are long the spread at  $-0.90$ . To take profit you sell it back later — at a higher (less negative) price.

## 1 You are long the spread

entry @  $-0.90$

## 2 March strengthens vs June

spread now =  $78.70 - 79.10 = -0.40$

## = Sell the spread to close

profit =  $-0.40 - (-0.90) = +0.50$

YOU MADE

**$+0.50$**

*per barrel*

≈ +\$500 per 1,000-bbl contract — you sold the spread back at its higher value.

*Illustrative prices, \$/barrel. A long spread profits when the spread rises.*

## THE PROBLEM

# Two liquid legs — but no market in the spread

You want to trade the March–June spread. Both outright books are full of resting orders — yet the spread's own book is empty. There is simply nothing to trade against.

### MARCH (real)

| PRICE | SIZE |
|-------|------|
| 100.5 | 8    |
| 100   | 12   |
| 99    | 10   |
| 98.5  | 14   |

### JUNE (real)

| PRICE | SIZE |
|-------|------|
| 98.5  | 11   |
| 98    | 18   |
| 97    | 15   |
| 96.5  | 9    |

### MARCH–JUNE SPREAD

– no asks –

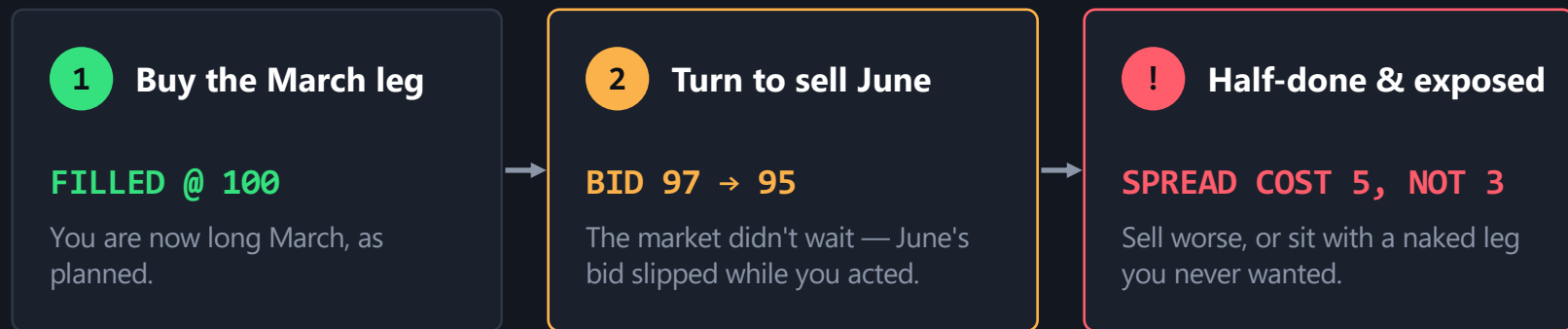
– no bids –

**EMPTY**

- Same barrels, same traders — but the liquidity is locked inside the two leg books. To anyone looking only at the spread, the market looks dead.

# Your only option: leg in by hand — and hope

With no market in the spread, you must send two separate orders and fill them one at a time. In between the fills, prices can move against you.

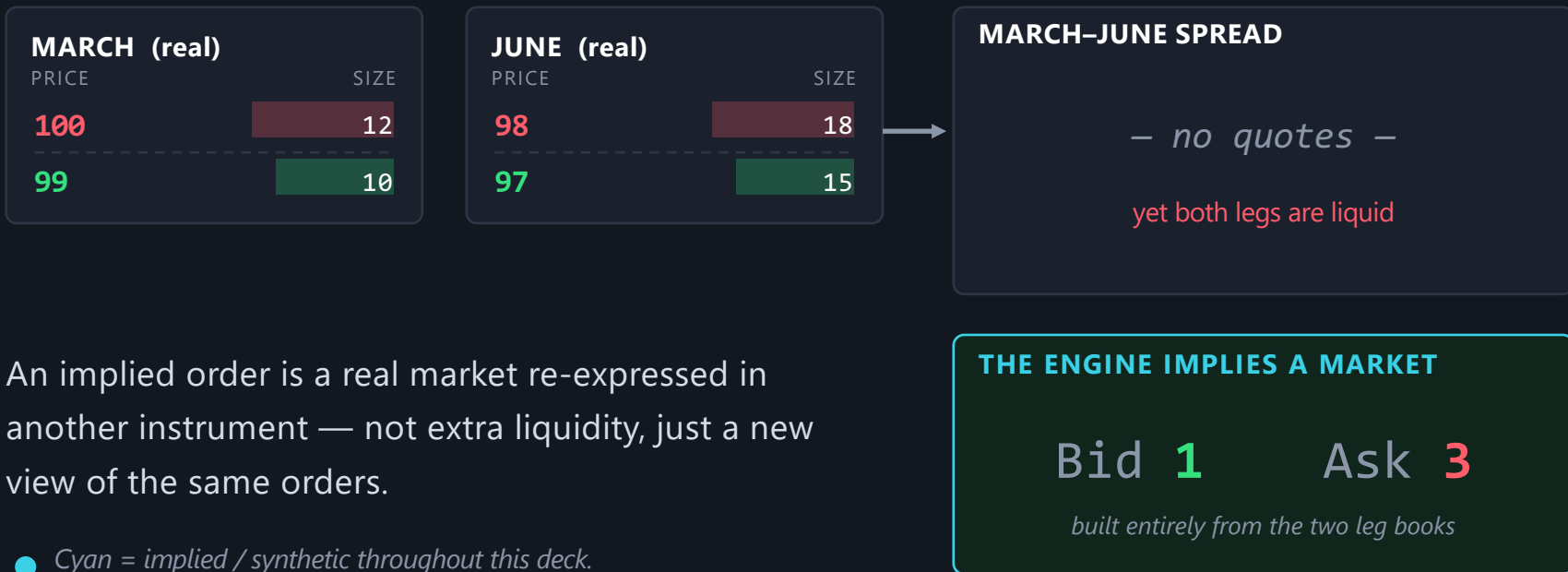


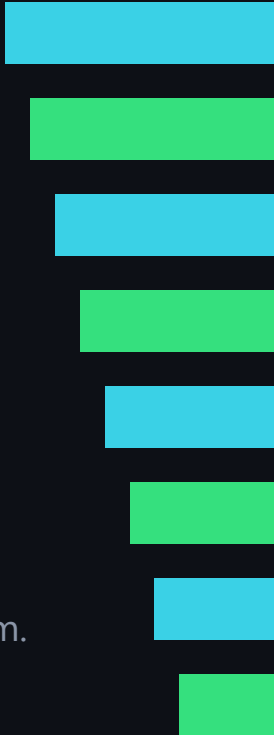
## LEGGING RISK

You can end up with one leg done and the other away from you — a worse spread, or an open position you never intended to hold.

## THE IDEA IN 30 SECONDS

# Liquidity you can't see — until the engine connects it





## SECTION I

# Foundations

The order book, outright instruments, and the spreads that are built on top of them.

# The order book: where buyers and sellers queue

## A SINGLE INSTRUMENT

| PRICE | SIZE |
|-------|------|
| 101   | 30   |
| 100   | 12   |
| 99    | 10   |
| 98    | 25   |

### ASKS — people willing to SELL

Lowest ask is the best price to buy at.

### BIDS — people willing to BUY

Highest bid is the best price to sell at.

Each row is a price level; the number is how much size is resting there. Orders match when a buyer and seller agree on price — highest bid meets lowest ask.

# An “outright” is just one thing, bought or sold

## EXAMPLE

### March Crude Future

One contract. You either buy it or sell it. Its price stands on its own — hence “outright.”

**BUY** March    **SELL** March

*Two directions, one instrument. Simple.*

## MARCH — LIVE BOOK

| PRICE | SIZE |
|-------|------|
| 101   | 30   |
| 100   | 12   |
| 99    | 10   |
| 98    | 25   |

Best market: buy at 100, sell at 99.

# Best bid, best ask, and the bid–ask spread

## MARCH

PRICE

SIZE

100

12

99

10

**BEST ASK (buy here)**

100

**BEST BID (sell here)**

99

**BID–ASK SPREAD** $100 - 99 = 1$ 

The gap a taker pays to cross. Tighter is better — and implied orders help tighten it.

# A combination instrument: the calendar spread

A spread is ONE tradeable instrument made of TWO legs. Buying it does both trades at once, in fixed proportion.

## DEFINITION

**Spread = March - June**

### BUY the spread

= BUY March + SELL June

### SELL the spread

= SELL March + BUY June

## ONE PRICE, TWO LEGS



# Why traders quote spreads directly



## Roll positions

Move exposure from an expiring month to the next in a single trade.



## Trade relationships

Bet on the gap between months, not the outright direction of the market.



## Lower leg risk

Both legs fill together, so you never get “half-done” at a bad price.

*Because spreads are traded in their own right, they need their own order book — and that is exactly where liquidity gets stranded.*

# The fragmentation problem

Each instrument has its own book. A spread trader and two outright traders can be willing to do exactly matching business — and never see each other.

## MARCH book

deep & liquid



walled off

## JUNE book

deep & liquid



walled off

## SPREAD book

thin / empty



walled off

## SECTION II

# The problem, made concrete

A trade that clearly should happen — but the books can't find each other on their own.

# Scenario: two liquid legs, an empty spread

| MARCH |      |
|-------|------|
| PRICE | SIZE |
| 100   | 12   |
| 101   | 30   |
| <hr/> |      |
| 99    | 10   |
| 98    | 25   |

| JUNE  |      |
|-------|------|
| PRICE | SIZE |
| 98    | 18   |
| 99    | 22   |
| <hr/> |      |
| 97    | 15   |
| 96    | 20   |

## MARCH-JUNE SPREAD

empty

no bids · no asks

A customer wants to BUY the spread at a fair price — but the spread book shows nothing. To them, the market looks dead.

# The liquidity is right there

Look what the leg books already allow, if we combine them:

## SELL the spread synthetically

**A** Sell March into its best bid  
fill @ 99

**B** Buy June from its best ask  
fill @ 98

---

$$\text{net spread} = 99 - 98 = 1$$

## SO THE SPREAD IS SELLABLE AT 1

The willingness to trade already exists — parked in the March and June books. Nothing new is needed.

The only missing piece:

**someone to post that price in the spread book automatically.**

## The question that implied orders answer

**“If the orders needed to make a market already exist in related books, why should the trader have to find and combine them by hand?”**

**Answer: they shouldn't. The matching engine does it for them — continuously, automatically.**

*It publishes IMPLIED ORDERS: synthetic quotes derived from real ones in other instruments.*

# Meet the implied order

## DEFINITION

An implied order is a synthetic bid or ask the engine posts in one instrument, derived from real resting orders in related instruments — and executable only by trading those real orders.

### ● Real-backed

Every implied order points to real orders standing behind it.

### ● Automatic

Recalculated live as the underlying books change.

### ● Disappears

If the real orders vanish, the implied one instantly does too.

### SECTION III

# Implied IN

Outright leg orders → a synthetic quote in the spread book.

# Setup: the two leg books we'll use

## MARCH

PRICE

SIZE

|       |    |
|-------|----|
| 100   | 12 |
| 101   | 30 |
| <hr/> |    |
| 99    | 10 |
| 98    | 25 |

## JUNE

PRICE

SIZE

|       |    |
|-------|----|
| 98    | 18 |
| 99    | 22 |
| <hr/> |    |
| 97    | 15 |
| 96    | 20 |

## BEST PRICES

March bid

99

March ask

100

June bid

97

June ask

98

*Spread = March – June. We'll imply its bid and ask next.*

# Build the implied spread BID

An implied BID = the best price at which the spread can be SOLD, using the legs.

1 To sell the spread, SELL March → hit the March bid

March bid = 99

2 ...and BUY June → lift the June ask

June ask = 98

3 implied spread bid =  $99 - 98 = 1$

IMPLIED SPREAD BID

1

= MarchBid - JuneAsk

Size =  $\min(10, 18) = 10$

# Build the implied spread ASK

An implied ASK = the best price at which the spread can be BOUGHT, using the legs.

- 1 To buy the spread, BUY March → lift the March ask

March ask = 100

- 2 ...and SELL June → hit the June bid

June bid = 97

- 3 implied spread ask =  $100 - 97 = 3$

IMPLIED SPREAD ASK

3

= MarchAsk - JuneBid

Size =  $\min(30, 15) = 15$

# A spread market appears out of the legs

Before: the spread book was empty. After: a live two-sided market — entirely implied.

BEFORE

MARCH–JUNE SPREAD

— empty —



AFTER (IMPLIED)

ASK

3

size 15

BID

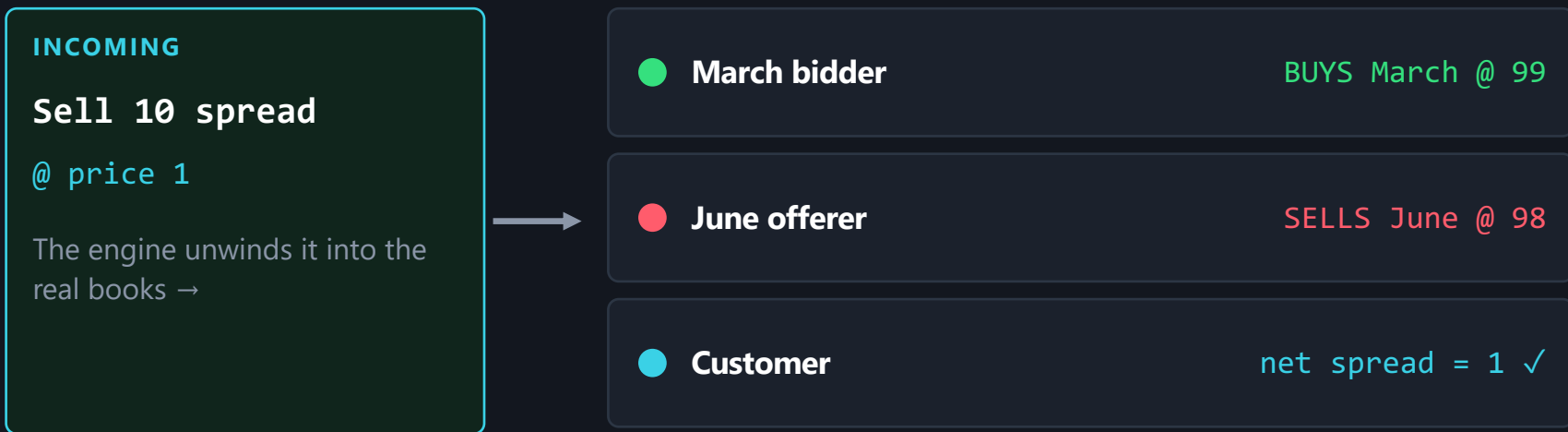
1

size 10

*Every price here is backed by real March & June orders.*

# Watch it trade: one order, three fills

A customer SELLS 10 of the spread at the implied bid of 1. Here's what actually executes:



*Three parties, one instant. The implied bid was just a convenient face on orders that were already there.*

# Implied IN, in one line

**real leg orders** → **implied spread quote**

*"IN" because liquidity flows INTO the combined instrument from its legs.*

**implied spread BID**

= March bid - June ask

**implied spread ASK**

= March ask - June bid

## SECTION IV

# Implied OUT

A spread order + one real leg → a synthetic quote in the other outright.

# Setup: a resting spread order meets a leg

Now the spread book is NOT empty. Someone is bidding the spread — and we'll turn that into an outright quote.

## SPREAD (real order)

**BID 2**

size 10

buy March, sell June

net 2

### JUNE

PRICE

SIZE

98

18

97

15

### MARCH

PRICE

SIZE

101

30

98

8

March bid is only 98 — weak. Watch the implied bid improve it.

# Build the implied MARCH bid

The spread buyer must BUY March and SELL June. Give the “sell June” leg to a real June buyer, and what's left is a March bid.

- 1 Spread buyer wants net  
March - June = 2
- 2 Real June buyer takes the June leg  
sell June @ 97 (June bid)
- 3 implied March bid = 2 + 97 = 99

## IMPLIED MARCH BID

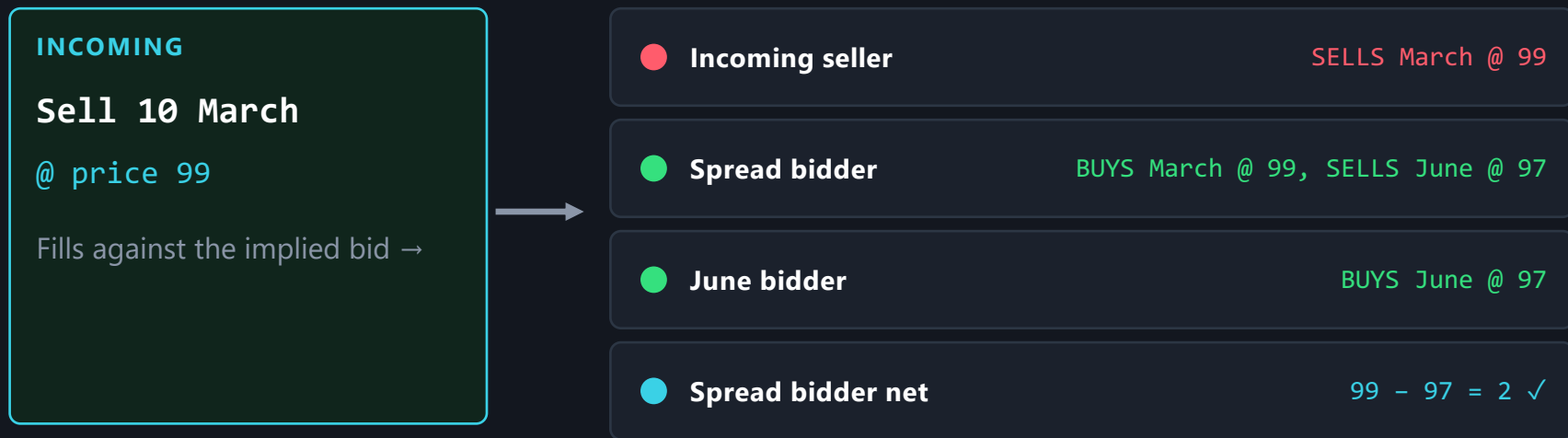
99

was 98 → now 99

The synthetic bid betters the real book.

# Watch it trade: the synthetic leg fills

A seller hits the implied March bid of 99 with 10 lots. Everyone gets exactly the trade they posted:



*The spread order and the June order already existed — the implied March bid simply let a March seller reach them.*

# The four building blocks

All implication reduces to arithmetic on best prices. For  $\text{Spread} = A - B$ :

IN

**Implied spread BID** $A \text{ bid} - B \text{ ask}$ 

IN

**Implied spread ASK** $A \text{ ask} - B \text{ bid}$ 

OUT

**Implied A bid** $\text{Spread bid} + B \text{ bid}$ 

OUT

**Implied B ask** $A \text{ ask} - \text{Spread bid}$ 

*Same principle each time: reshuffle real orders across instruments so a price is guaranteed to be executable.*

## Implied OUT, in one line

spread order + real leg → implied outright  
quote

*"OUT" because liquidity flows OUT of the combined instrument into an outright leg.*

**IN and OUT are two directions of the same bridge. Together they keep every related book in sync — automatically.**

## SECTION V

# Why it's so powerful

In combination, implication makes markets tighter, deeper-looking, and more likely to fill.

# Tighter markets and better fills

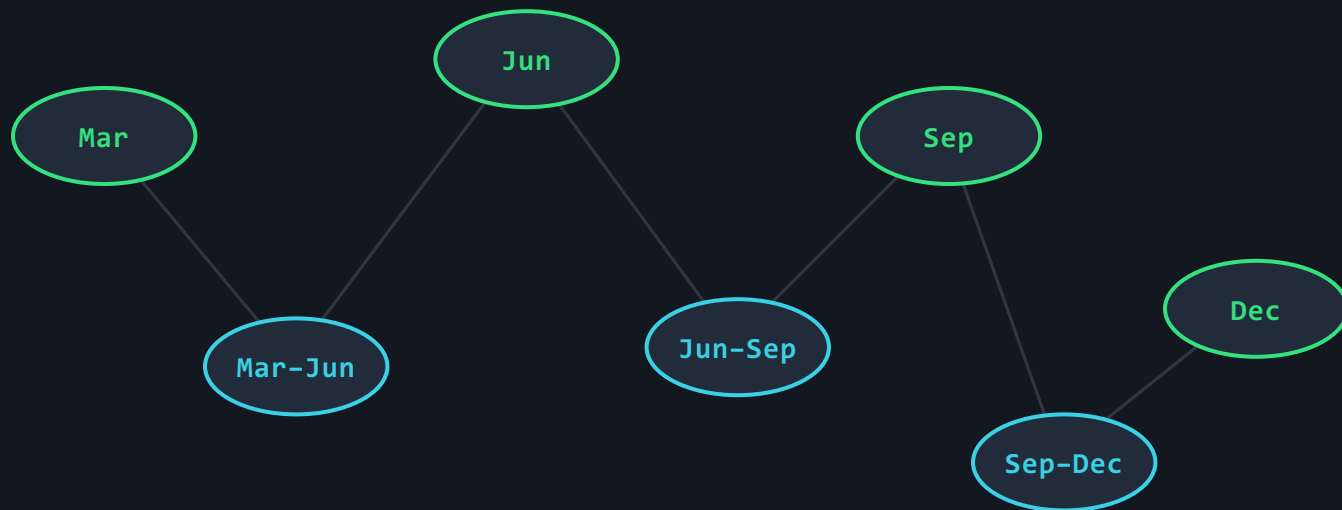
Add implied quotes to a book and the best bid/ask can only stay the same or improve.



*Same underlying orders — a 3-wide market becomes 1-wide. Takers pay less to cross.*

# One trade, a whole web of linked books

In a strip of months, every outright and every spread implies into the others — a connected mesh.



*Green = outright · Cyan = spread. A single new leg order can refresh implied prices across many books at once.*

# Chaining: implied-from-implied

An implied order can itself feed the next implication — a chain across instruments.



Each extra level unlocks prices no single book could show — but also multiplies the work the engine must do. That trade-off is Section VII.

## SECTION VI

# No liquidity is created

The single most important thing to understand: implication re-expresses liquidity, it never invents it.

# The big misconception

## THE MYTH

**“Implied orders conjure extra liquidity out of thin air, so the market is deeper than the real orders justify.”**

**✗ False.**

## THE REALITY

**Every implied order is just real orders wearing a different label. Trade the implied and you trade the reals — once.**

**✓ Re-expression, not creation.**

# Every implied is backed 1:1 by real orders

Peel back any implied quote and you find specific real orders standing behind it.



*No hidden inventory. The implied size is capped by the smallest real order behind it (here, 10).*

# Pull the real order, the implied vanishes

Implied quotes are not stored liquidity — they are live calculations. Remove an input and the output disappears instantly.

## BEFORE

March bid 99 × 10 ✓

June ask 98 × 18 ✓

---

Implied spread bid @ 1

live ✓

## AFTER (March bid cancels)

March bid — gone ✕

June ask 98 × 18 ✓

---

Implied spread bid

vanishes ✕

# No double-counting when it trades

The engine guarantees each real order is consumed at most once — whether it's hit directly or through an implied.

**March bid 99 can be reached TWO ways:**

directly (someone sells March) · or via the implied spread bid

## THE RULE

The moment it fills either way, it is removed from the book — so its liquidity is counted exactly once. Implication reshuffles who trades with whom; it never lets the same order trade twice.

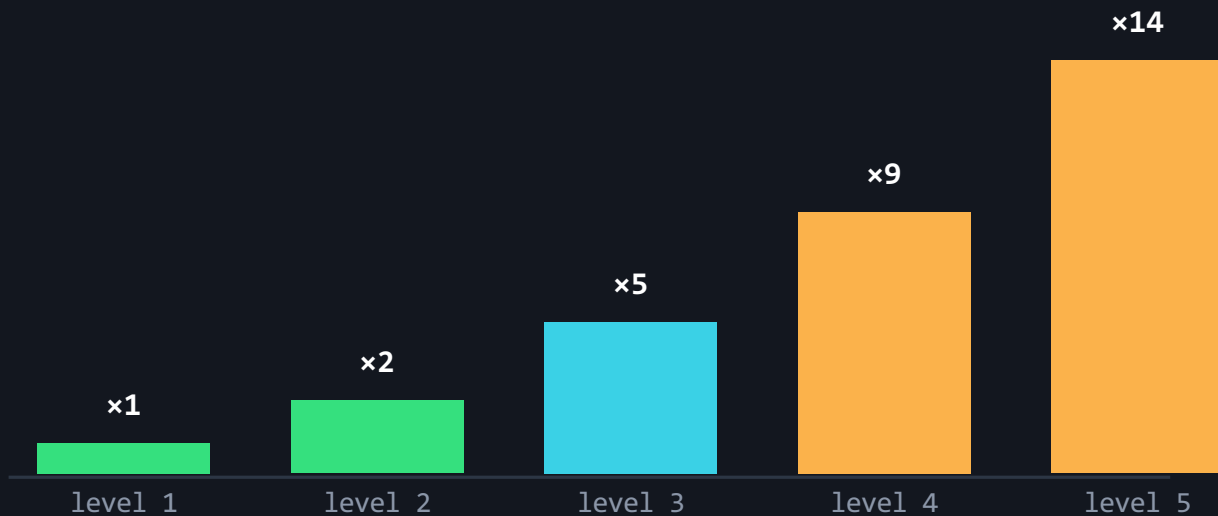
## SECTION VII

# The cost of going deep

Why more levels of implication explode the workload — and why real platforms cap them.

# Combinatorial explosion

Each added level of implication multiplies the combinations the engine must compute and keep current.



## GROWTH

Combinations rise faster than linearly with depth — quickly into the thousands for a busy strip.

# What deep implication actually costs



## Latency

Every book change can trigger a cascade of re-calculations before the next order is processed.



## Compute

Thousands of potential implieds must be generated, priced, and ranked continuously.



## Complexity

Deeper chains are harder to audit, risk-check, and reason about when something breaks.

*The engine has microseconds. Unbounded implication depth would blow that budget.*

# Why real platforms cap the levels

## THE ENGINEERING CHOICE

Exchanges limit implication to a small number of levels (often 1–2, sometimes a few more) — capturing most of the benefit for a fraction of the cost.

Level 1–2



most of the tightening

Level 3–4



sharply diminishing returns

Level 5+



rarely worth the latency

**The cap is a deliberate trade-off: near-all the liquidity benefit, bounded and predictable latency.**

# Key takeaways

## ● Implied orders connect books

Synthetic quotes let outright and spread traders reach each other automatically.

## ● IN and OUT are one bridge

Legs imply spreads (IN); spreads + a leg imply outright (OUT).

## ● No liquidity is created

Every implied is backed 1:1 by real orders and vanishes when they do.

## ● Depth is capped on purpose

Combinatorial cost forces platforms to limit implication levels.

# Quick glossary

## Outright

A single instrument, bought or sold on its own.

## Spread / combo

One instrument defined as a combination of legs, e.g. March – June.

## Best bid / ask

Highest price to sell at / lowest price to buy at.

## Implied IN

Spread quote derived from the outright leg books.

## Implied OUT

Outright quote derived from a spread order plus a real leg.

## Implication depth

How many chained levels of implication the engine allows.

**You now have the full picture: what implied orders are, how they're built, why they help, and why they're bounded.**